THE OCEANS: CURRENTS, TIDES, SALINITY / INDIAN OCEAN AND IOD

OCEAN-CURRENT FORCE CHAIN → DENSITY-OVERTURNING CHAIN → GYRE COMPARISON → UPWELLING-FOOD WEB → TIDE GEOMETRY → LOCAL TIDAL RANGE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 OCEAN-CURRENT FORCE CHAIN

OCEAN-CURRENT FORCE CHAIN | WIND STRESS | SURFACE MOMENTUM | TURNING BY HEMISPHERE | PRESSURE GRADIENT | GEOSTROPHIC TENDENCY | BASIN + FRICTION | REALISED CURRENT PATH

WIND STRESS → surface momentum → CORIOLIS → turning by hemisphere → PRESSURE GRADIENT → geostrophic tendency → BASIN + FRICTION → realised current path → MUST REMEMBER: Ocean structure links temperature, salinity and density to stratification and circulation

CORE CLAIM

- WIND STRESS → surface momentum
- CORIOLIS → turning by hemisphere

MECHANISM

- PRESSURE GRADIENT → geostrophic tendency
- BASIN + FRICTION → realised current path

CONSEQUENCE / LIMIT

- MUST REMEMBER: Ocean structure links temperature, salinity and density to stratification and circulation
- wind stress, Coriolis, Ekman transport, gyres, upwelling, thermohaline overturning, tides and Indian Ocean monsoon reversal culminate in IOD–ENSO

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Ocean structure links temperature, salinity and density to stratification and circulation; wind stress, Coriolis, Ekman transport, gyres, upwelling, thermohaline overturning, tides and Indian Ocean monsoon reversal culminate in IOD–ENSO interactions.

01

STAGE 01 DENSITY-OVERTURNING CHAIN

DENSITY-OVERTURNING CHAIN | DENSITY INCREASE | DENSITY DECREASE | DENSE WATER | SINKING AND SPREADING | MIXING + TOPOGRAPHY | OVERTURNING PATHWAYS

COOLING / EVAPORATION → density increase → FRESHWATER / WARMING → density decrease → DENSE WATER → sinking and spreading → MIXING + TOPOGRAPHY → overturning pathways

CORE CLAIM

- COOLING / EVAPORATION → density increase
- FRESHWATER / WARMING → density decrease

MECHANISM

- DENSE WATER → sinking and spreading
- MIXING + TOPOGRAPHY → overturning pathways

CONSEQUENCE / LIMIT

- COOLING / EVAPORATION → density increase
- FRESHWATER / WARMING → density decrease

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DENSE WATER → sinking and spreading; MIXING + TOPOGRAPHY → overturning pathways.

02

STAGE 02 GYRE COMPARISON

GYRE COMPARISON | NORTH SUBTROPICS | CLOCKWISE TENDENCY | SOUTH SUBTROPICS | ANTICLOCKWISE TENDENCY | WESTERN LIMB | NARROW, SWIFT, WARM | EASTERN LIMB

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
NORTH SUBTROPICS → clockwise tendency	WESTERN LIMB → narrow, swift, warm	NORTH SUBTROPICS → clockwise tendency
SOUTH SUBTROPICS → anticlockwise tendency	EASTERN LIMB → broad, cool, upwelling-prone	SOUTH SUBTROPICS → anticlockwise tendency

CORE CLAIM

- NORTH SUBTROPICS → clockwise tendency
- SOUTH SUBTROPICS → anticlockwise tendency

MECHANISM

- WESTERN LIMB → narrow, swift, warm
- EASTERN LIMB → broad, cool, upwelling-prone

CONSEQUENCE / LIMIT

- NORTH SUBTROPICS → clockwise tendency
- SOUTH SUBTROPICS → anticlockwise tendency

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WESTERN LIMB → narrow, swift, warm; EASTERN LIMB → broad, cool, upwelling-prone.

03

STAGE 03 UPWELLING-FOOD WEB

UPWELLING-FOOD WEB | ALONGSHORE WIND | EKMAN TRANSPORT OFFSHORE | DEEPER WATER RISES | NUTRIENT SUPPLY | OXYGEN, LIGHT, HABITAT, FISHING PRESSURE

ALONGSHORE WIND → Ekman transport offshore → DEEPER WATER RISES → nutrient supply → PHYTOPLANKTON → zooplankton → fish → LIMITS → oxygen, light, habitat, fishing pressure

CORE CLAIM

- ALONGSHORE WIND → Ekman transport offshore
- DEEPER WATER RISES → nutrient supply

MECHANISM

- PHYTOPLANKTON → zooplankton → fish
- LIMITS → oxygen, light, habitat, fishing pressure

CONSEQUENCE / LIMIT

- ALONGSHORE WIND → Ekman transport offshore
- DEEPER WATER RISES → nutrient supply

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WESTERN LIMB → narrow, swift, warm; EASTERN LIMB → broad, cool, upwelling-prone.

03

STAGE 03

UPWELLING-FOOD WEB

UPWELLING-FOOD WEB · ALONGSHORE WIND · EKMAN TRANSPORT OFFSHORE · DEEPER WATER RISES · NUTRIENT SUPPLY · OXYGEN, LIGHT, HABITAT, FISHING PRESSURE

ALONGSHORE WIND → Ekman transport offshore → DEEPER WATER RISES → nutrient supply → PHYTOPLANKTON → zooplankton → fish → LIMITS → oxygen, light, habitat, fishing pressure

CORE CLAIM

- ALONGSHORE WIND → Ekman transport offshore
- DEEPER WATER RISES → nutrient supply

MECHANISM

- PHYTOPLANKTON → zooplankton → fish
- LIMITS → oxygen, light, habitat, fishing pressure

CONSEQUENCE / LIMIT

- ALONGSHORE WIND → Ekman transport offshore
- DEEPER WATER RISES → nutrient supply

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PHYTOPLANKTON → zooplankton → fish; LIMITS → oxygen, light, habitat, fishing pressure.

04

STAGE 04

TIDE GEOMETRY

TIDE GEOMETRY · FULL MOON · ALIGNED FORCES · SPRING TIDE · LARGER RANGE · QUARTER MOON · PARTIAL OFFSET · NEAP TIDE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
NEW / FULL MOON → aligned forces	QUARTER MOON → partial offset	NEW / FULL MOON → aligned forces
SPRING TIDE → larger range	NEAP TIDE → smaller range	SPRING TIDE → larger range

CORE CLAIM

- NEW / FULL MOON → aligned forces
- SPRING TIDE → larger range

MECHANISM

- QUARTER MOON → partial offset
- NEAP TIDE → smaller range

CONSEQUENCE / LIMIT

- NEW / FULL MOON → aligned forces
- SPRING TIDE → larger range

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUARTER MOON → partial offset; NEAP TIDE → smaller range.

05

STAGE 05

LOCAL TIDAL RANGE

LOCAL TIDAL RANGE · ASTRONOMICAL FORCING · INCOMING TIDE · SHELF DEPTH + BAY SHAPE · RESONANCE + FRICTION · TIMING AND HEIGHT · SAME FORCING, DIFFERENT LOCAL RANGE

ASTRONOMICAL FORCING → incoming tide → SHELF DEPTH + BAY SHAPE → amplification → RESONANCE + FRICTION → timing and height → RULE → same forcing, different local range

CORE CLAIM

- ASTRONOMICAL FORCING → incoming tide
- SHELF DEPTH + BAY SHAPE → amplification

MECHANISM

- RESONANCE + FRICTION → timing and height
- RULE → same forcing, different local range

CONSEQUENCE / LIMIT

- ASTRONOMICAL FORCING → incoming tide
- SHELF DEPTH + BAY SHAPE → amplification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESONANCE + FRICTION → timing and height; RULE → same forcing, different local range.

06

STAGE 06

SALINITY BUDGET

SALINITY BUDGET · EVAPORATION + BRINE · SALINITY RISES · RAIN + RIVER + MELT · SALINITY FALLS · ADVECTION + MIXING · REDISTRIBUTE SALT · TEMP + SALINITY

EVAPORATION + BRINE → salinity rises

RAIN + RIVER + MELT → salinity falls → ADVECTION + MIXING → redistribute salt → TEMP + SALINITY → density and stratification

CORE CLAIM

- EVAPORATION + BRINE → salinity rises
- RAIN + RIVER + MELT → salinity falls

MECHANISM

- ADVECTION + MIXING → redistribute salt
- TEMP + SALINITY → density and stratification

CONSEQUENCE / LIMIT

- EVAPORATION + BRINE → salinity rises
- RAIN + RIVER + MELT → salinity falls

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RAIN + RIVER + MELT → salinity falls.

SALINITY BUDGET · EVAPORATION + BRINE · SALINITY RISES · RAIN + RIVER + MELT · SALINITY FALLS · ADVECTION + MIXING · REDISTRIBUTE SALT · TEMP + SALINITY

EVAPORATION + BRINE → salinity rises

RAIN + RIVER + MELT → salinity falls → ADVECTION + MIXING → redistribute salt → TEMP + SALINITY → density and stratification

CORE CLAIM

- EVAPORATION + BRINE → salinity rises
- RAIN + RIVER + MELT → salinity falls

MECHANISM

- ADVECTION + MIXING → redistribute salt
- TEMP + SALINITY → density and stratification

CONSEQUENCE / LIMIT

- EVAPORATION + BRINE → salinity rises
- RAIN + RIVER + MELT → salinity falls

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RAIN + RIVER + MELT → salinity falls.

07

STAGE 07 ARABIAN-BAY CONTRAST

ARABIAN-BAY CONTRAST · ARABIAN SEA · STRONGER EVAPORATION · BAY OF BENGAL · RAIN + LARGE RIVER INPUT · GENERAL RESULT · ARABIAN SEA MORE SALINE · SEASON AND LOCAL MIXING MATTER

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
ARABIAN SEA → stronger evaporation	GENERAL RESULT → Arabian Sea more saline	ARABIAN SEA → stronger evaporation
BAY OF BENGAL → rain + large river input	QUALIFIER → season and local mixing matter	BAY OF BENGAL → rain + large river input

CORE CLAIM

- ARABIAN SEA → stronger evaporation
- BAY OF BENGAL → rain + large river input

MECHANISM

- GENERAL RESULT → Arabian Sea more saline
- QUALIFIER → season and local mixing matter

CONSEQUENCE / LIMIT

- ARABIAN SEA → stronger evaporation
- BAY OF BENGAL → rain + large river input

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BAY OF BENGAL → rain + large river input.

08

STAGE 08 MONSOON-CURRENT SWITCH

MONSOON-CURRENT SWITCH · SW MONSOON · WINDS DRIVE SUMMER REORGANISATION · SOMALI SECTOR · STRONG CURRENT + UPWELLING · NE MONSOON · NORTHERN FLOWS REVERSE/REORGANISE · MAP RULE

SW MONSOON → winds drive summer reorganisation

SOMALI SECTOR → strong current + upwelling → NE MONSOON → northern flows reverse/reorganise → MAP RULE → label every arrow by season

CORE CLAIM

- SW MONSOON → winds drive summer reorganisation
- SOMALI SECTOR → strong current + upwelling

MECHANISM

- NE MONSOON → northern flows reverse/reorganise
- MAP RULE → label every arrow by season

CONSEQUENCE / LIMIT

- SW MONSOON → winds drive summer reorganisation
- SOMALI SECTOR → strong current + upwelling

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MAP RULE → label every arrow by season.

09

STAGE 09 IOD COUPLED SEESAW

IOD COUPLED SEESAW · WEST-EAST SST CONTRAST · PRESSURE GRADIENT · THERMOCLINE AND UPWELLING RESPONSE · RAINFALL SHIFTS ZONALLY · INDIA EFFECT · CONDITIONAL ON TIMING + ENSO + MJO · CLOSE DISTINCTION

WEST-EAST SST CONTRAST → pressure gradient

WINDS → thermocline and upwelling response → CONVECTION → rainfall shifts zonally → INDIA EFFECT → conditional on timing + ENSO + MJO

CORE CLAIM

- WEST-EAST SST CONTRAST → pressure gradient
- WINDS → thermocline and upwelling response

MECHANISM

- CONVECTION → rainfall shifts zonally
- INDIA EFFECT → conditional on timing + ENSO + MJO

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Wave transmits energy, current transports water and tide is astronomical sea-level oscillation
- spring/neap are alignment effects, not seasons, and positive IOD is a west-minus-east Indian Ocean SST gradient rather than

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Wave transmits energy, current transports water and tide is astronomical sea-level oscillation; spring/neap are alignment effects, not seasons, and positive IOD is a west-minus-east Indian Ocean SST gradient rather than simply a warm western ocean.

WINDS → thermocline and upwelling response

CONVECTION → rainfall shifts zonally

INDIA EFFECT → conditional on timing + ENSO + MJO

CORE CLAIM

- WEST-EAST SST CONTRAST → pressure gradient
- WINDS → thermocline and upwelling response

MECHANISM

- CONVECTION → rainfall shifts zonally
- INDIA EFFECT → conditional on timing + ENSO + MJO

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Wave transmits energy, current transports water and tide is astronomical sea-level oscillation
- spring/neap are alignment effects, not seasons, and positive IOD is a west-minus-east Indian Ocean SST gradient rather than

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Wave transmits energy, current transports water and tide is astronomical sea-level oscillation; spring/neap are alignment effects, not seasons, and positive IOD is a west-minus-east Indian Ocean SST gradient rather than simply a warm western ocean.

10

STAGE 10 OBSERVATION-TO-SERVICE CHAIN

OBSERVATION-TO-SERVICE CHAIN · OCEAN + WEATHER OBSERVATIONS · DATA MANAGEMENT AND PRODUCTS · ANALYSIS AND FORECAST ENSEMBLES · MARINE SERVICE WITH UNCERTAINTY · EVIDENCE LIMIT · IOD AND ENSO MODULATE PROBABILITIES RATHER THAN DETERMINE INDIAN · EVENT PHASE, SEASON, INTERACTION, BASIN BACKGROUND AND MODEL/SOURCE DATE

OMNI / RAMA → ocean + weather observations

INCOIS → data management and products

MODELS → analysis and forecast ensembles

USER → marine service with uncertainty

CORE CLAIM

- OMNI / RAMA → ocean + weather observations
- INCOIS → data management and products

MECHANISM

- MODELS → analysis and forecast ensembles
- USER → marine service with uncertainty

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: IOD and ENSO modulate probabilities rather than determine Indian rainfall
- event phase, season, interaction, basin background and model/source date must qualify fisheries, cyclone or monsoon conclusions.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: event phase, season, interaction, basin background and model/source date must qualify fisheries, cyclone or monsoon conclusions.

11

STAGE 11 OCEAN ANSWER SPINE

OCEAN ANSWER SPINE · DEPTH, SCALE AND SEASON · FORCE, DENSITY OR GRAVITY · BASIN, CURRENT, TIDE OR SALINITY · VARIABILITY, ECOLOGY AND EVIDENCE DATE

DEFINE → depth, scale and season

DERIVE → force, density or gravity

MAP → basin, current, tide or salinity

QUALIFY → variability, ecology and evidence date

CORE CLAIM

- DEFINE → depth, scale and season
- DERIVE → force, density or gravity

MECHANISM

- MAP → basin, current, tide or salinity
- QUALIFY → variability, ecology and evidence date

CONSEQUENCE / LIMIT

- DEFINE → depth, scale and season
- DERIVE → force, density or gravity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MAP → basin, current, tide or salinity; QUALIFY → variability, ecology and evidence date.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · THE ARABIAN SEA HOSTS ONE OF THE WORLD'S STRONGEST · WHY HERE · MONSOON-DRIVEN PRODUCTIVITY AND ORGANIC-MATTER DECAY COMBINE WITH WEAK VENTILATION. · LOSS OF FIXED NITROGEN VIA DENITRIFICATION/ANAMMOX , RELEASE OF · FISHERY SHIFTS, AND REPLACEMENT OF DIATOMS BY HYPOXIA-TOLERANT NOCTILUCA · MAJID HUSAIN LINKS ABNORMALLY HIGH SEA TEMPERATURES WITH CORAL · MARINE HEATWAVE

OPTIONAL SOURCE DEPTH

- CAUTION: The Arabian Sea hosts one of the world's strongest tropical oxygen-minimum zones at
- CAUTION: Why here: monsoon-driven productivity and organic-matter decay combine with weak ventilation.
- CAUTION: Consequences: loss of fixed nitrogen via denitrification/anammox , release of the greenhouse gas nitrous oxide (N_2O) ,
- fishery shifts, and replacement of diatoms by hypoxia-tolerant Noctiluca (green Noctiluca blooms) — an India-relevant fisheries/climate concern.

ADVANCED DEBATE

- Majid Husain links abnormally high sea temperatures with coral bleaching, where heat stress kills the microscopic algae that colour and feed corals.
- CAUTION: Marine heatwave = a prolonged spell of unusually high sea-surface temperature
- over a specific ocean region, relative to its normal seasonal range.
- CAUTION: Mechanism: ocean warming + weak winds/mixing + persistent high pressure can suppress cooling and keep surface waters anomalously warm.

LEGACY / NUANCE

- CAUTION: India link: Arabian Sea and Bay of Bengal warming can intensify cyclones, stress corals and alter fisheries/monsoon moisture supply.
- North Indian Ocean currents reverse seasonally with the monsoon.
- Summer = SW Monsoon Drift; winter = NE Monsoon Drift.
- South Indian Ocean has a normal subtropical gyre.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels